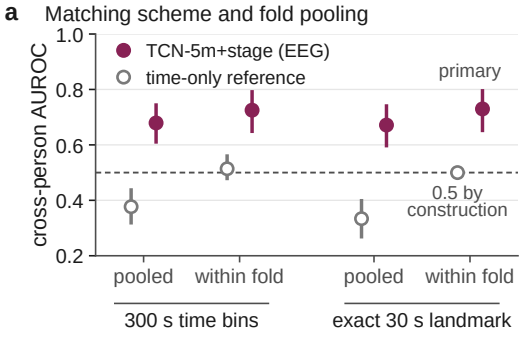
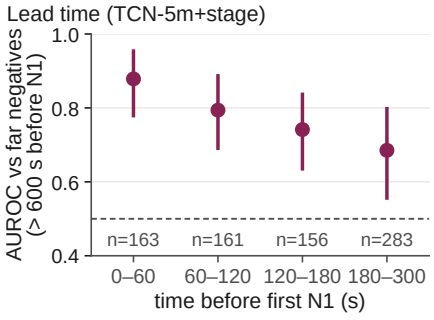
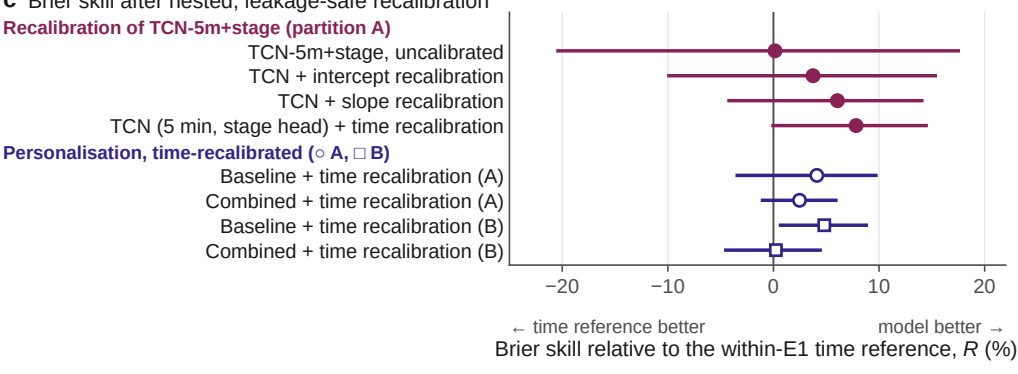

a
Matching scheme and fold pooling
cross-person AUROC
1.0
0.8
0.6
0.4
0.2
TCN-5m+stage (EEG)
time-only reference
primary
0.5 by construction
pooled
within fold
pooled
within fold
300 s time bins
exact 30 s landmark


b
Lead time (TCN-5m+stage)
AUROC vs far negatives (> 600 s before N1)
1.0
0.8
0.6
0.4
n=163
n=161
n=156
n=283
0–60
60–120
120–180
180–300
time before first N1 (s)


c
Brier skill after nested, leakage-safe recalibration
Recalibration of TCN-5m+stage (partition A)
TCN-5m+stage, uncalibrated
TCN + intercept recalibration
TCN + slope recalibration
TCN (5 min, stage head) + time recalibration
Personalisation, time-recalibrated (○ A, □ B)
Baseline + time recalibration (A)
Combined + time recalibration (A)
Baseline + time recalibration (B)
Combined + time recalibration (B)
−20
−10
0
10
20
← time reference better
model better →
Brier skill relative to the within-E1 time reference, R (%)